

Math552 - Lab 1

2026-08-28

Jason Tinant

Objectives

- Learn the basic Jupyter Notebook workflow.
- Load data into a pandas DataFrame.
- Estimate simple probabilities from real data.
- Create a simple graph.

Verify setup

1. Open a Jupyter notebook using the shortcut you created.
2. In a blank Python cell, run the following code to prove to me that your Python is working.

```
import sys
import os

print(sys.executable)
print(os.getcwd())
```

```
/Users/cjtinant/PROJECTS/ThinkBayes2/.venv/bin/python
/Users/cjtinant/PROJECTS/ThinkBayes2/math552_labs
```

3. Load the General Social Survey data and display the first five observations. Before loading the data, check (from above) your working directory. If your working directory is Math 452_552 use:

```
import pandas as pd

gss = pd.read_csv("../data/gss_bayes.csv", index_col=0)
gss.head()
```

	year	age	sex	polviews	partyid	indus10
caseid						
1	1974	21.0	1	4.0	2.0	4970.0
2	1974	41.0	1	5.0	0.0	9160.0
5	1974	58.0	2	6.0	1.0	2670.0
6	1974	30.0	1	5.0	4.0	6870.0
7	1974	48.0	1	5.0	4.0	7860.0

4. Run the pandas command `gss.info()`. Explain the output.

Data Exploration

Run each command in a separate Python cell.

```
gss.columns
```

```
count    49290.000000
mean      46.143132
std       17.111420
min       18.000000
25%       32.000000
50%       44.000000
75%       59.000000
max       89.000000
Name: age, dtype: float64
```

```
gss.shape
```

```
gss.describe()
```

```
gss['sex'].value_counts()
```

```
gss['age'].describe()
```

1. How many variables are there?
2. What does describe() tell you?
3. Is sex numeric or categorical? Explain.
4. What does value_counts() do?
5. Run the command gss.isna().sum(). Which variables contain missing values?

Probability Estimation

- Estimate the probability that a randomly selected respondent is female.

```
p_female = (gss['sex'] == 2).mean()
print(p_female)
```

0.5378575776019476

- Why does .mean() compute a probability?
- Estimate the probability a respondent is over age 50.
- Estimate the probability that a randomly selected respondent is female and over age 50.
- What is the probability that someone is over 50 given they are female? (Hint: this is a conditional probability)
- Is the previous probability the same as the probability that someone is female given they are over 50? Explain.
- What is the probability that someone is over 50 given they are male?

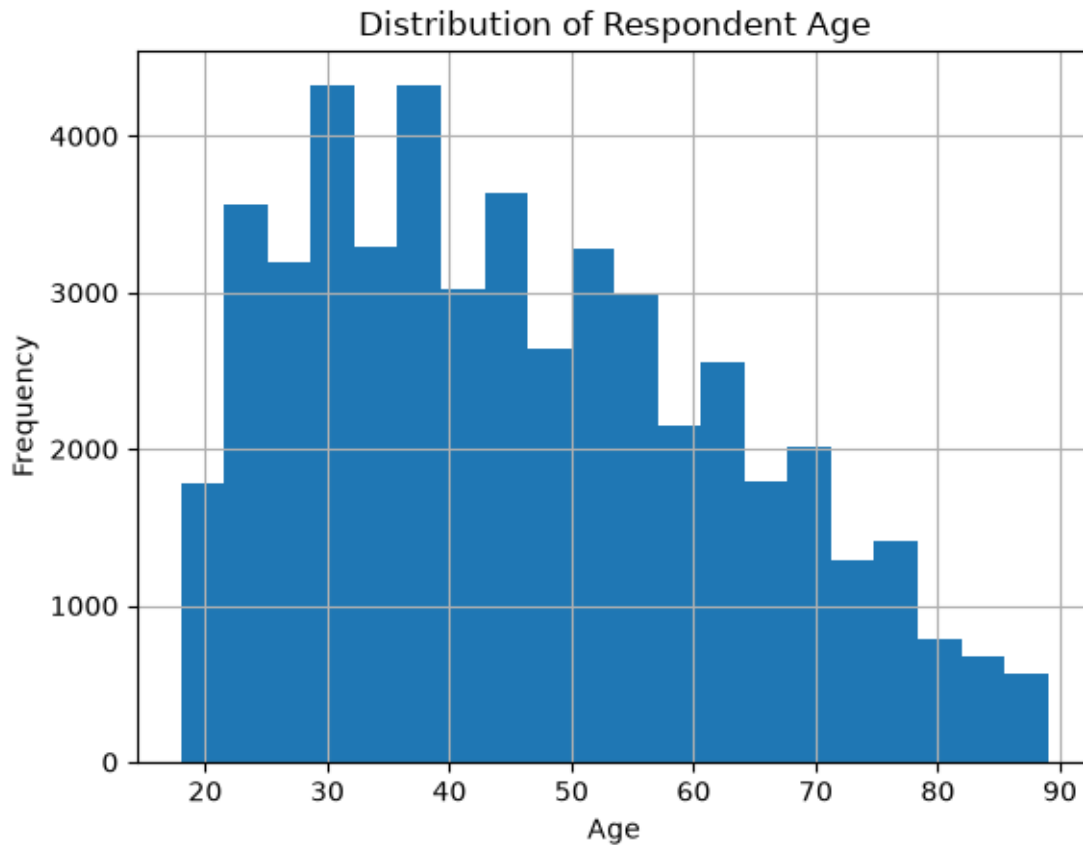
Plotting

- Create a histogram of age.

```
import matplotlib.pyplot as plt

gss["age"].hist(bins=20)

plt.xlabel("Age")
plt.ylabel("Frequency")
plt.title("Distribution of Respondent Age")
plt.show()
```



- Describe the shape of the age distribution.
- Is it symmetric? Skewed? Are there any unusual features?

Reflection

In two or three sentences, describe one thing you learned about Jupyter Notebook and one thing you learned about estimating probabilities from data.

Deliverable

After completing this lab, generate a .pdf file and upload it to the dropbox on D2L.